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Project Title:	Smart Therapeutic Ultrasound Device for Mission-Critical Medical Care		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
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Zip Code:	98105-6606	**Congressional District:**	7
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
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Task Description:

(1) The original aims of project.
The principal, long-term objective of this proposed effort is to develop a smart medical device that would be lightweight, portable, FDA approved, under commercial development, and capable of addressing the various aspects of the four risks described above. In particular, we seek to address under Risk 43, the risk of blunt internal trauma and internal bleeding resulting, for example, from a collision in microgravity between an astronaut and a heavy object. The device we propose would be capable of detecting internal bleeding and inducing transcutaneous hemostasis. Under Risks 23 and 38, as recent experiences in Antarctica demonstrate [Nielsen, 2001], malignant tumors that require some form of surgery may well appear without warning, even when extensive pre-screening is undertaken. The device we propose would be capable of performing bloodless ablation of detectable tumors without fear of metastasis. Under Risk 12, the formation of renal calculi has been suggested by Schneider, et al. [1994] and by Ball and Evans [2001] as a likely result in extended space flight; indeed, a precedent exists of early mission termination due to a ureteral stone [Pietrzyk, et al, 2003]. The device we propose would be capable of detecting and sufficiently comminuting a renal calculus for it to pass without obstruction through the urinary system. Figure 1 shows a cartoon describing these various concepts and a conceptual prototype of our proposed smart medical device.

Specific Aim 1. To expand the capability of our existing smart medical device for transcutaneous acoustic hemostasis to include a fully integrated ultrasound detection, targeting, and therapy system (CRL 5-6).

Specific Aim 2. To develop the capability of the integrated image-guided therapy system in SA 1 to target and ablate specific tissue volumes such as benign and malignant tumors (CRL 4-5).

Specific Aim 3. To develop the capability of the integrated image-guided therapy system in SA 1 to target and comminute renal calculi (CRL 3-4). (2) the key findings of the project In the initial grant and through other grants, we have developed many high intensity focused ultrasound (HIFU) systems to test stopping bleeding, necrosing tissues, and comminuting kidney stones. A broad array of successful specific applications within these areas have been published. These devices continue to be refined to make them easier to use and acceptable for the FDA approval. Primarily this first year of this grant, efforts to significantly enhance efficacy have been investigated. Primarily these efforts are to accelerate tissue heating with HIFU. This will enable stopping bleeding in larger vessels and treating cancer tumors completely in a reasonable amount of time. The NSBRI funded research has led to the following main findings that are elaborated in the claims section and in the publications listed in this report. -Two frequencies can accelerate heating and hemostasis. -Addition of micro-bubble contrast agents can accelerate heating and hemostasis. -Nonlinear acoustic propagation can be exploited to accelerate heating. -A model was developed to account for heating by cavitation bubbles. -A model was developed to optimize heating enhancement through nonlinear acoustics. -An industrial collaboration has been established to engineer high-power amplifiers to affect the methods listed above. -A high power system was used to complete full thickness coagulation in organs and to occlude vessels as large as 1 cm in diameter. In addition, in this year progress was made in the study of HIFU to comminute kidney stones, in efforts to guide treatment, and in moving toward clinical trials.

The main findings are listed here. -High power, 2 frequency systems have proven effective in eroding kidney stones. -A model was developed and tested to predict kidney stone fracture. -A response to the first review of our patent application to synchronize imaging and HIFU has been submitted. -A GE-funded Chinese clinical HIFU machine will be tested in the NSBRI lab before being moved to the radiology department for cancer treatments in humans and animals.

(3) the impact of these findings on the hypotheses or requirements (technology), objectives and specific aims of the original proposal These results make it possible to deliver in a predictable way the energy required for therapeutic effect. We are working toward stopping major bleeds that could not otherwise be stopped. We have imported technology used to treat cancer patients in China. And we are working with collaborators to advance renal stone comminution with HIFU.

(4) the proposed research plan for the coming year. The proposed research plan for the coming year is to stay with the timetable proposed in the grant application. Careful in vitro experiments compared to modeling will be continued to define precisely the roles of cavitation and nonlinear acoustics in enhancing heating and to optimize these effects. This work includes an investigation of how cavitation bubbles grow. The same techniques will be applied to measuring the acoustic and cavitation fields of the clinical HIFU device. In vivo testing will continue with high power devices, micro-bubbles, and dual frequency systems to enhance nonlinear acoustics and cavitation. The goal is to seal large vessels. A student from our Japanese collaborator group will join our group starting January 2006 and we will determine optimal frequency and pulsing regimes to break kidney stones using both in vitro experiments and numerical models.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

We have proposed to develop technology to use high intensity focused ultrasound (HIFU) to arrest bleeding, necrotize cancer, and comminute renal cacluli in space. These risks to astronauts are enumerated and ranked in the bioastronautics critical path roadmap, but solutions to these 3 medical ailments is also of great importance on Earth. Occult bleeding is also a major Earth-based risk. In a multi-center review of 72,151 trauma admissions at a civilian health center, 62% of those requiring emergent operative treatment were in shock on arrival in the operating room, and the actual cause of death was credited to bleeding in 82% of the cases [Hoyt, et al., 1994]. Indeed, exsanguination is often reported as the major cause of mortality (~50%) in combat casualties [Zajtchuk and Sullivan, 1995]. Accordingly, we have been undertaking a major effort, funded by a variety of federal agencies (but principally by the DoD) to develop a portable device that could be used in forward echelons of the battlefield to address the risk of death by exsanguination. We have leveraged this extensive DoD-funded effort in our research for the NSBRI, and propose its continued development under Specific Aim 1. This image-guided therapy approach utilizes the capability of HIFU to rapidly elevate the temperature to coagulative necrosis levels (> 55oC), thus inducing hemostasis. The capability to induce coagulative necrosis, particularly under image-guidance, provides a unique opportunity to treat benign and malignant tumors. Our laboratory has been active in the development of this technology and recently hosted an international conference [Andrew, Vaezy, and Crum, 2003] that addressed in some detail the many international efforts to use HIFU to treat cancer. The image-guided, transcutaneous, acoustic hemostasis device that we have proposed under Specific Aim 1 can also be utilized to target and ablate tumors, and this is our second Specific Aim. Our laboratory has had considerable experience in the treatment of stone disease; we are now participating in the 10th year of an NIH Program Project Grant to study the

physical basis of stone comminution and tissue damage in shock-wave lithotripsy. Between 5% and 15% of the general population will develop a urinary calculus during their lifetime [Consensus conference. JAMA 1988;260:977] and will account for about 7 to 10 of every 1000-hospital admissions in the United States [Levy et al. 1995]. One of our major contributions to lithotripsy research has been the determination that acoustic cavitation plays a major role in stone comminution [McAteer, et al., 2003; Pishchalnikov, et al., 2002; Sapozhnikov, et al., 2002; Evan, et al., 2002]; thus, we contend that our smart medical device, which can easily produce cavitation at the high intensities it is capable of generating, will be able to comminute a stone in the collecting system of the kidney. Our third specific aim is to undertake R&D to demonstrate this capability of our SMS device.
The NSBRI funded research is developing medical technology to stop bleeding, kill cancers, and break up kidney stones. The technology is high intensity focused ultrasound (HIFU), where by sound waves are focused through the skin for therapeutic effect deep in the body. Several companies are working with our group to produce clinical machines and some HIFU machines are FDA approved. In addition NSBRI funding has contributed to education. Several students from Electrical Engineering, Mechanical Engineering, Bioengineering, Physics, Applied Math, and even high school have worked on this project as part of their degree. In addition a course was taught to Industrial Design students challenging them to conceptualize HIFU devices. We have also hosted Japanese, Russian, and French students as part of collborations built through NSBRI funding.

Task Progress:

Tasks 1a-d: To develop a probe applicator containing an embedded image probe together with an annular array capable of delivering HIFU at therapy levels...To integrate the probe applicator with a human interface and power unit... To perform a series of experiments in ex vivo ... to induce acoustic hemostasis transcutaneously. An integrated applicator has been developed and tested. It is not possible to insert a figure here but the device is decribed in the manuscript by Owen and in the patent application. Preliminary work has begun on an array system.
2a-c. To develop a real-time guidance control and targeting system that permits HIFU lesions to be produced at specific sites ... To develop a lesion-imaging technique to permit real-time monitoring of the HIFU lesion produced ... of interest in ex vivo and phantom tissues. These 3 tasks as well as in vivo application have been completed with a simple image-guidance system. The system relies on a bright region on B-mode or Doppler imaging to indicate the HIFU focus. This is the image-guidance technique that is used the clinical system we are testing and our patent application describes a method and a device to permit imaging while using HIFU. More sophisticated imaging techniques are being developed. We recently graduated a student who developed a model-based approached to monitor temperature in the tissue during HIFU based on detected changes in sound speed. This system has also completed the 3 tasks but has not been developed into a real-time system. Temperature monitoring is beign developed for the clinical machien as well.

2d: To develop a therapy protocol based upon a validated numerical model... A model based on linear acosutic propagation was developed in FEMlab for use in temperature monitoring described above. A more refined model incorporating nonlinear acoustic propagation and written in finite difference is being tested versus in vtro measurements.

3a-c. To determine through both modeling and in vitro experimentation the acoustic protocols required to produce cavitation levels sufficient to comminute artificial kidney stones... To measure the acoustic pressure amplitudes and intensities that are produced... To determine the minimal sizes of renal calculi that can be detected... Dr. Bailey worked for 1 month in Japan with our collaborators on using their system to comminute stones. We have now obtained and replicated much of their system. A student from the Japanaese will join our group in January 2006. Our French colleagues are also advising and will visit for th emonth of August 2005. We have used a fiber optic probe hydrophone to measure precisely th epressures up to 80 MPa, and pasive cavitation detectors and high speed camera to observe cavitation. We are imaing different stones in tissue and gels using B-mode ultrasound and vibroacosutography.

3d. To demonstrate the capability of the integrated system...An integrated system for comminution has not yet been tested.

Two invention disclosures in the patenting process were listed. One is a method ot synchronize the HIFU with an ultrasound imager. The response to the reiview fo the first patent application has just been filed. Two is the use of 2 frequencies to accelerrate hemostasis. The patent application still has not been filed. We have two pending proposals to develop new technology. One is to use HIFU to control bleedign during partal nephrectomy. This is with a prominent urologist Ralph Clayman. Two is to utilize soem new techniques of applying HIFU to stop bleeding in teh extremeties with Engineerign Acoustics Inc.

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Awards	Maxwell A. "Adam Maxwell won the Acoustical Society of America (ASA) Student Papaer Award, Spring 2005." May-2005
Awards	Bailey M. "Michael Bailey was elected Technical Committee Chair for Biomedical Ultrasound at the ASA, 2004." Jan-2004
Awards	Vaezy S. "Shahram Vaezy was promoted to Assoc. Prof. of Bioengineering at the Univ. of Washington, 2004." Jan-2004
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